

|                                                          |                                               |
|----------------------------------------------------------|-----------------------------------------------|
| Program : <b>Diploma in Cloud Computing and Big Data</b> |                                               |
| Course Code : <b>3278</b>                                | Course Title: Linux System Administration Lab |
| Semester : <b>3</b>                                      | Credits: <b>2.5</b>                           |
| Course Category: <b>Engineering Science</b>              |                                               |
| Periods per week: <b>4 (L: 1 T: 0 P: 3)</b>              | Periods per semester: <b>60</b>               |

### Course Objectives:

- Familiarize the students with the basics of Operating System Installation and configuration
- To understand basic linux text editors
- To Understand basic linux commands and shell scripting

### Course Prerequisites:

| Topic                               | Course Name                    |
|-------------------------------------|--------------------------------|
| Basic knowledge in Computer systems | Introduction to IT Systems Lab |

### Course Outcomes:

On completion of the course student will be able to:

| COn | Description                                                                                                         | Duration (Hours) | Cognitive Level |
|-----|---------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Demonstrate the installation and configuration of Operating Systems, basic files and directory management commands. | 11               | Applying        |
| CO2 | Experiment with Linux Package and Process management tools                                                          | 17               | Applying        |
| CO3 | Make use of Linux text editors and experiment with text processing commands                                         | 12               | Applying        |

|     |                                                                |    |          |
|-----|----------------------------------------------------------------|----|----------|
| CO4 | Utilize Shell scripting to automate linux administrative tasks | 14 | Applying |
|     | Lab Test                                                       | 6  |          |

### CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     | 3   |     |     |     |
| CO2             | 2   |     |     |     |     |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             | 2   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline

| Module Outcomes | Name of Experiment                                                                                                                                                      | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Demonstrate the Installation of Operating Systems , and demonstrate basic file and directory management commands.</b>                                                |                  |                 |
| M1.01           | Install and configure different operating systems.                                                                                                                      | 3                | Applying        |
| M1.02           | Illustrate Basic networking configuration of a linux machine                                                                                                            | 2                | Applying        |
| M1.03           | Demonstrate Basic file and directory management commands (ls, dir, pwd, mkdir, chmod, cp, mv, rm, rmdir, cat, find, tar). Familiarize linux documentation and man pages | 6                | Applying        |
| <b>CO2</b>      | <b>Experiment with Linux Package and Process management tools</b>                                                                                                       |                  |                 |
| M2.01           | Understand file system structure of linux (types of files, file permissions).                                                                                           | 3                | Understanding   |

|            |                                                                                                  |   |          |
|------------|--------------------------------------------------------------------------------------------------|---|----------|
| M2.02      | Demonstrate package management tools and installation of application (apt-get or yum)            | 2 | Applying |
| M2.03      | Illustrate Process and system statistics using proc files, ps, free, kill, top, htop, atop, vtop | 8 | Applying |
| M2.04      | Demonstrate remote system administration using ssh command and file transfer using scp command   | 4 | Applying |
| <b>CO3</b> | <b>Make use of Linux text editors and experiment with text processing commands</b>               |   |          |
| M3.01      | Experiment with UNIX editors such as vi, ed, ex and EMACS, nano                                  | 4 | Applying |
| M3.02      | Illustrate cut, grep, awk, sort, uniq, head, tail commands and output redirection using pipe( )  | 8 | Applying |
|            | Lab Test – I                                                                                     | 3 |          |
| <b>C04</b> | <b>Utilize Shell scripting to automate linux administrative tasks</b>                            |   |          |
| M4.01      | Demonstrate branching statements                                                                 | 2 | Applying |
| M4.02      | Illustrate looping statements                                                                    | 4 | Applying |
| M4.03      | Apply cron job to schedule task                                                                  | 4 | Applying |
| M4.04      | Discover with Open Ended Experiments**                                                           | 4 | Applying |
|            | Lab Test – II                                                                                    | 3 |          |

**\*\* - Sample Open Ended Experiment**

1. Schedule a task to perform a daily backup of certain databases and files to cloud storage or a physical server
2. Schedule a task that monitors the free space of the system hard disk. If the free space drops below the threshold, the user must be alerted with a message and current date

**Text / Reference**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                        |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| T1         | Essential System Administration: Tools and Techniques for Linux and Unix Administration, Third Edition by Leen Frisch ,O'Reilly |

**Online Resources**

| <b>Sl No</b> | <b>Website Link</b>                                                   |
|--------------|-----------------------------------------------------------------------|
| 1            | <a href="https://www.shellscript.sh/">https://www.shellscript.sh/</a> |